

The Sum Is Less Than Its Parts: Decomposing the Attention Write by Source Detects Contextual Hallucinations

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Abstract

When a language model answers from a retrieved document, its attention heads mix evidence from the context with evidence from the model’s own self-generated tokens and write the *sum* into the residual stream. We split each head’s write into these two source-specific contributions before they are summed, and find that the two sides are complementary. The *context* write, read out along the model’s unembedding, gives a training-free groundedness score (GROUNDHOG_U). The *response* write (its magnitude and directional agreement with the other residual writes) gives a supervised hallucination detector (GROUNDHOG). Across grounded-generation benchmarks and model families, GROUNDHOG beats prior internal-state detectors, and GROUNDHOG_U is the strongest training-free groundedness signal on the multi-model benchmarks.

1 Introduction

Retrieval-augmented generation conditions a model on an external source so its answers can be grounded in that text (Lewis et al., 2020); when they are not, the model produces *contextual hallucinations* unsupported by the source. A line of detectors looks *inside* the model, on the premise that grounding leaves a trace in the representations. Attention-based ones either track *where* a head attends (Lookback Lens, Chuang et al., 2024) or inspect *what* it finally writes after all attended positions are combined (NoVo, Ho et al., 2025; TPA, Lu et al., 2026); none preserve the source-specific write vectors before they are summed, which is exactly where we find the signal.

Inside an attention head, the value-weighted sum over attended positions can be partitioned by *source*: the retrieved context versus the model’s own generated history. These two writes carry *opposite* evidence (grounded generation writes from the context, while hallucination coincides with writing from the response history), so collapsing them

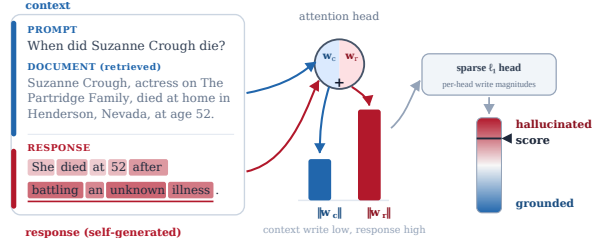


Figure 1: **GROUNDHOG**. Each head’s write splits by source into a context write w_c and a response write w_r . Reading w_c along the unembedding scores groundedness without training (GROUNDHOG_U); the response write’s size and its agreement with the other writes detects hallucination (GROUNDHOG). The unsplit sum cancels the two opposite signals.

into one head output cancels much of the signal (Figure 1). We use the sides for different tasks: the context write for *groundedness*, training-free (GROUNDHOG_U); the response write for *hallucination*, supervised (GROUNDHOG).

2 The Source Split

For attention head h at a generated position, let α_j^h be the attention weight to prior position j , v_j^h its value vector, and W_O^h the head’s slice of the output projection. We partition attended positions into a context set C (prompt and retrieved document) and a response-history set R (previously generated tokens), and restrict the value sum to each: $w_c^h = W_O^h \sum_{j \in C} \alpha_j^h v_j^h$ and $w_r^h = W_O^h \sum_{j \in R} \alpha_j^h v_j^h$, exact complements of the ordinary head write ($w^h = w_c^h + w_r^h$). A feed-forward write w_f^h completes the residual update.

Two oppositely-signed, independent streams

By each head’s write-magnitude AUROC (Figure 2), context writes are grounding-discriminative (AUROC<0.5) and response writes hallucination-discriminative (>0.5), while the *total* write piles at chance. Since $\|w_c + w_r\|^2 = \|w_c\|^2 + \|w_r\|^2 + 2 w_c^\top w_r$, the ordinary head output sums

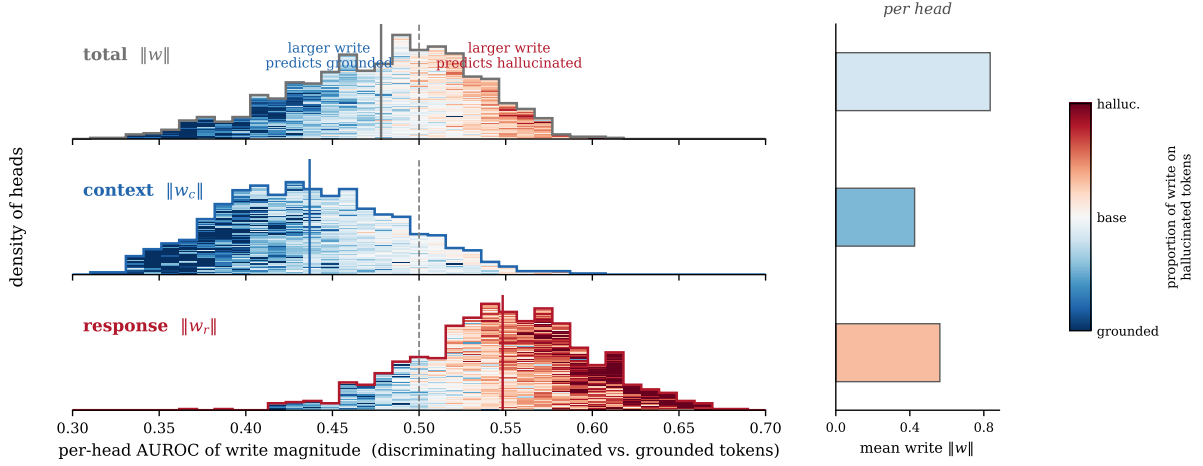


Figure 2: **The split, per head** (Llama-2-7B, HDM). *Left*: per-head *discrimination* (AUROC of the write magnitude, hallucinated vs. grounded tokens; 0.5=chance), each head colored by its *polarity* (fraction of its write mass on hallucinated tokens, centered on the base rate). Context is grounding-discriminative and low-polarity (blue), the response write the opposite (red), the total at chance; the two are distinct. *Right*: mean write per stream, colored by mean polarity.

two opposite-signed terms and cancels them; the split recovers both.

Context side: groundedness, training-free (GROUNDHOG_U) We read the context write along the emitted token’s unembedding direction u_y : $q^h = u_y^\top w_c^h$. Summed over heads and emitted tokens, this is an exact additive logit contribution from the retrieved context to what the model says. It requires no labels or fitting; for Table 2, low source contribution is oriented as higher risk.

Response side: hallucination, supervised (GROUNDHOG) We read the response write through three per-head quantities: its magnitude $\|w_r^h\|$, its agreement with the context write $\cos(w_c^h, w_r^h)$, and its agreement with the feed-forward write $\cos(w_r^h, w_f^h)$. Each is a separate sparse (ℓ_1) logistic readout over heads; a small ℓ_2 meta-model weights the three scores.

3 Results

Table 1 reports response-level AUROC. GROUNDHOG is the strongest detector on every benchmark, ahead of *steelmanned* baselines (competitors’ own per-head features read by our ℓ_1 head, improving on their published pipelines), so the gain comes from the source split. Table 2 reports the complementary context-side use: among same-model, training-free groundedness signals, GROUNDHOG_U has the best mean AUROC on the three multi-model benchmarks, significantly over the next best row on HDM and RAGTruth.

Method	HDM-ctx	RT	Ragog-ans	RB
models	3	3	3	3
TPA [†]	0.809	0.751	0.729	0.730
ReDeEP [†]	0.845	0.814	0.808	0.745
Lookback [†]	0.830	0.735	0.808	0.735
SAPLMA	0.870	0.807	0.830	0.797
NoVo [†]	0.856	0.823	0.850	0.795
GROUNDHOG	0.881***	0.842*	0.868***	0.813***

Table 1: **Supervised response-level hallucination detection AUROC** on HDM-context, RAGTruth, RAGog-nize answerable-only, and RAGBench (Paudel et al., 2025; Niu et al., 2024; Ridder et al., 2026; Friel et al., 2025), averaged over three model architectures (Llama2-7B, Mistral-7B, Qwen3-4B). GROUNDHOG is best on every benchmark. Significance level denoted by (*/**/*** : $p < 0.05/0.01/0.001$). [†]: *steelmanned* baselines (the method’s own per-head features read by our improved ℓ_1 regression protocol).

Training-free	HDM-ctx	RT	Ragog-ans
readers	3	3	3
GROUNDHOG _U (ours)	0.699***	0.647***	0.654
LUMINA	0.606	0.610	0.648
TPA-RAG	0.614	0.610	0.619
Entropy	0.602	0.605	0.631
Perplexity	0.629	0.620	0.626

Table 2: **Training-free token-level evaluation.** Token-level AUROC: hallucinated or unsupported tokens are ranked against supported tokens from the same response. Values are averaged over models for datasets with token-level labels. */**/*** marks GROUNDHOG_U vs. next-best at $p < 0.05/0.01/0.001$.

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